AC209B · MILESTONE 2

Does anyone write down *where they go home to?*

Housing documentation in discharge summaries for patients with serious mental illness

PROJECT

#63 · Data Processing & EDA

TEAM

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Why this project

When patients with serious mental illness leave the hospital, whether they have stable housing plays a huge role in whether they stay healthy or end up back in the hospital.

THE STARTING OBSERVATION

RESEARCH QUESTION

Is housing mentioned in SMI discharge summaries — and do documentation rates differ by race, gender, or insurance?

WHY IT MATTERS

- Unstable housing drives medication non-adherence and missed follow-ups
- Clinical guidelines say housing should be in the discharge plan
- Without it, discharge plans can fail before the patient leaves the hospital

Three datasets, three jobs

1

MIMIC-IV

Structured tables

- admissions, patients, diagnoses_icd
- Demographics, insurance, ICD codes
- 546,028 admissions

2

MIMIC-IV-Note

Clinical free text

- Discharge summaries (7 CSV files)
- Linked to admissions by hadm_id
- 331,793 notes

3

MIMIC-III-SDOH

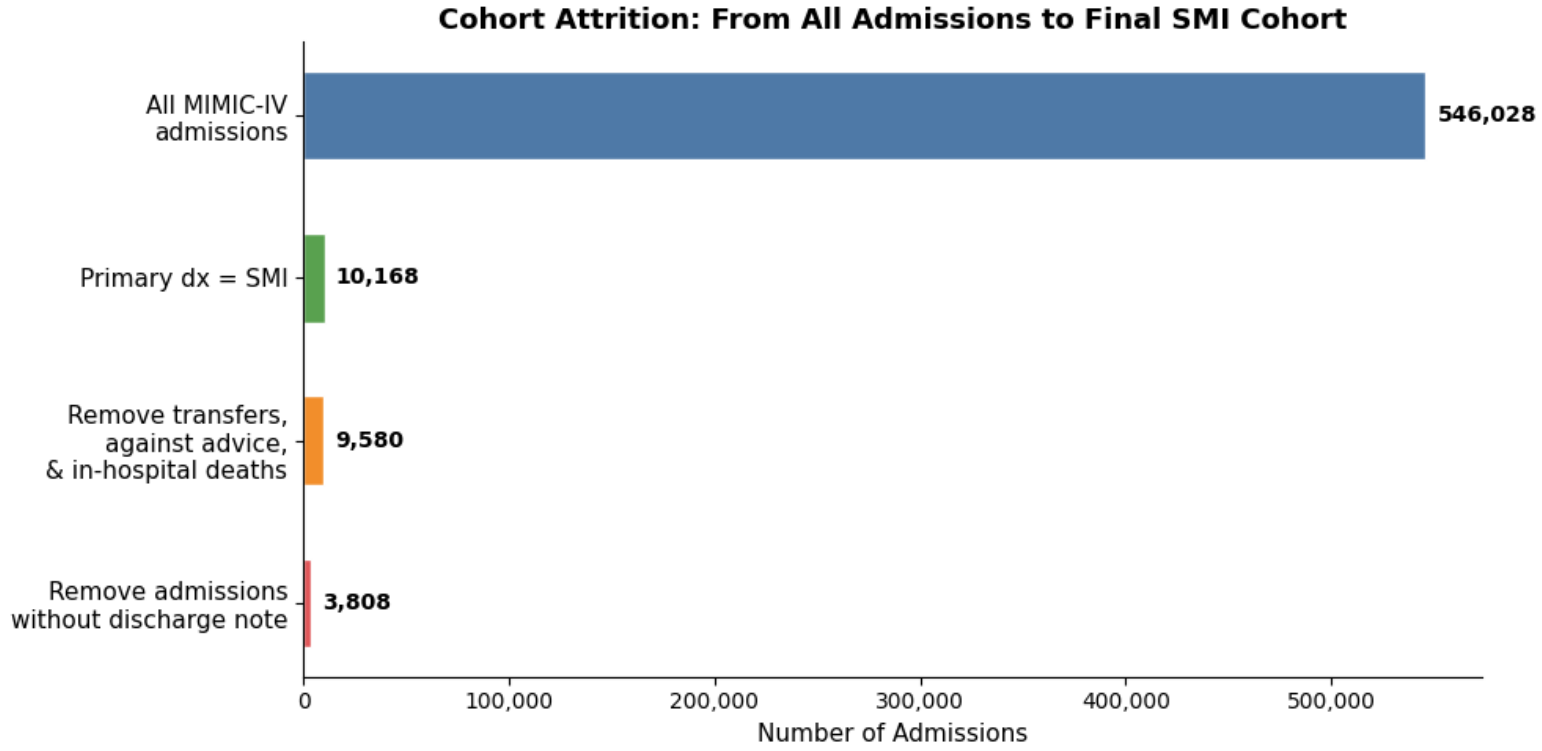
Validation set

- Hand-labeled clinical sentences
- Used to test our keyword search
- 1,800 labeled sentences

Building the cohort

1. **Merge**
   MIMIC-IV admissions + patients on subject_id
2. **Categorize**
   Race (4), insurance (4), gender — simplified groups
3. **Outcomes**
   30-day readmission flag (~20%); in-hospital death flag
4. **Diagnose**
   Attach first-listed ICD code per admission
5. **Filter**
   Keep only schizophrenia, bipolar, severe/recurrent MDD
6. **Exclude**
   Drop transfers, against-advice, in-hospital deaths
7. **Attach text**
   Merge MIMIC-IV-Note discharge summaries by hadm_id
8. **Clean**
   Strip the administrative “Discharge Disposition” line
9. **Search**
   Apply the validated 20-keyword housing search

From 546,028 admissions to 3,808



3,808

FINAL ADMISSIONS

after exclusions and note matching

2,859

UNIQUE PATIENTS

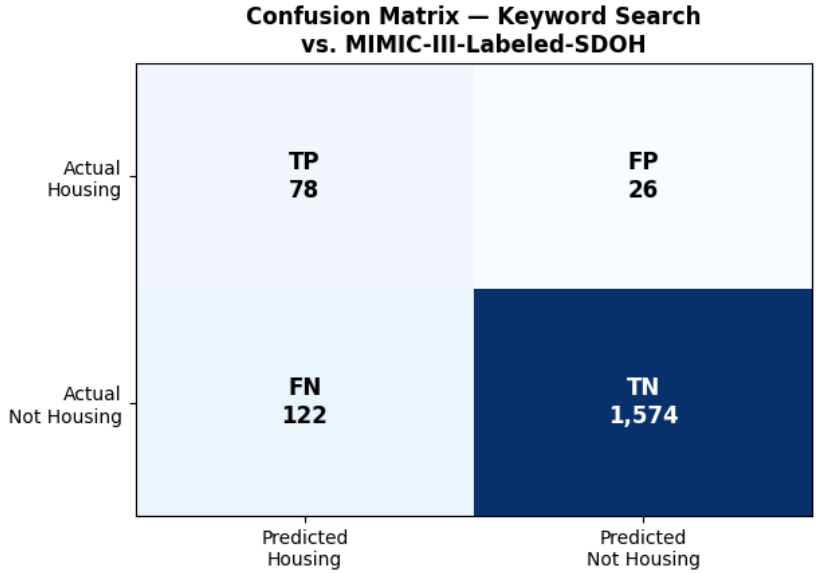
some patients have multiple stays

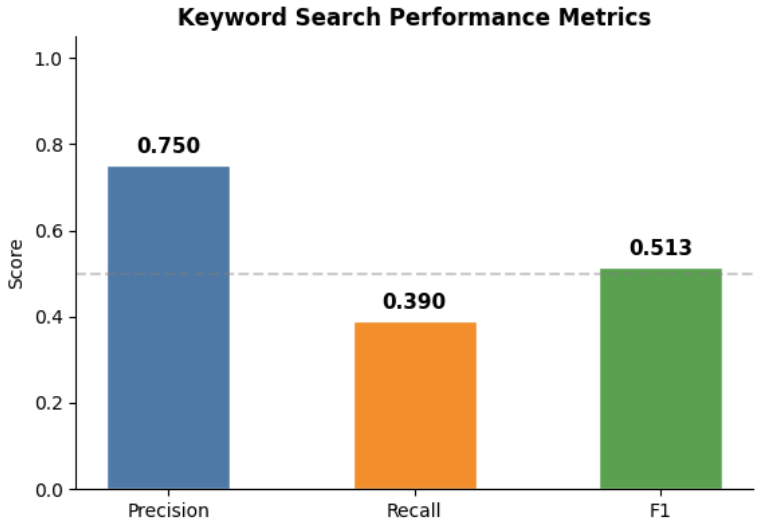
0.7%

RETENTION RATE

of all MIMIC-IV admissions

How good is keyword search, really?





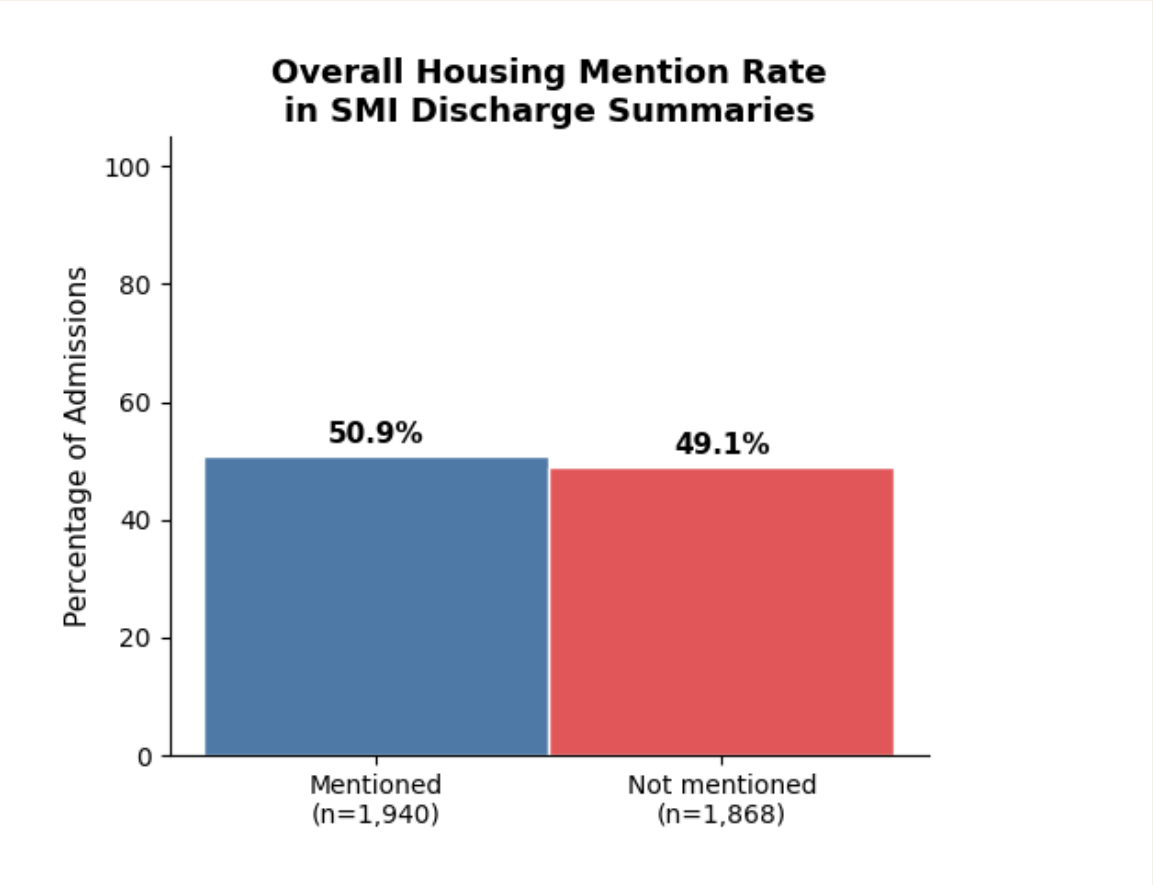
0.75
PRECISION

0.39
RECALL

0.51
F1 SCORE

When we flag housing we're usually right — but we miss 6 in 10 real mentions. Keyword search is a floor, not a ceiling. Motivates ClinicalBERT in Milestone 3.

Half the notes mention housing

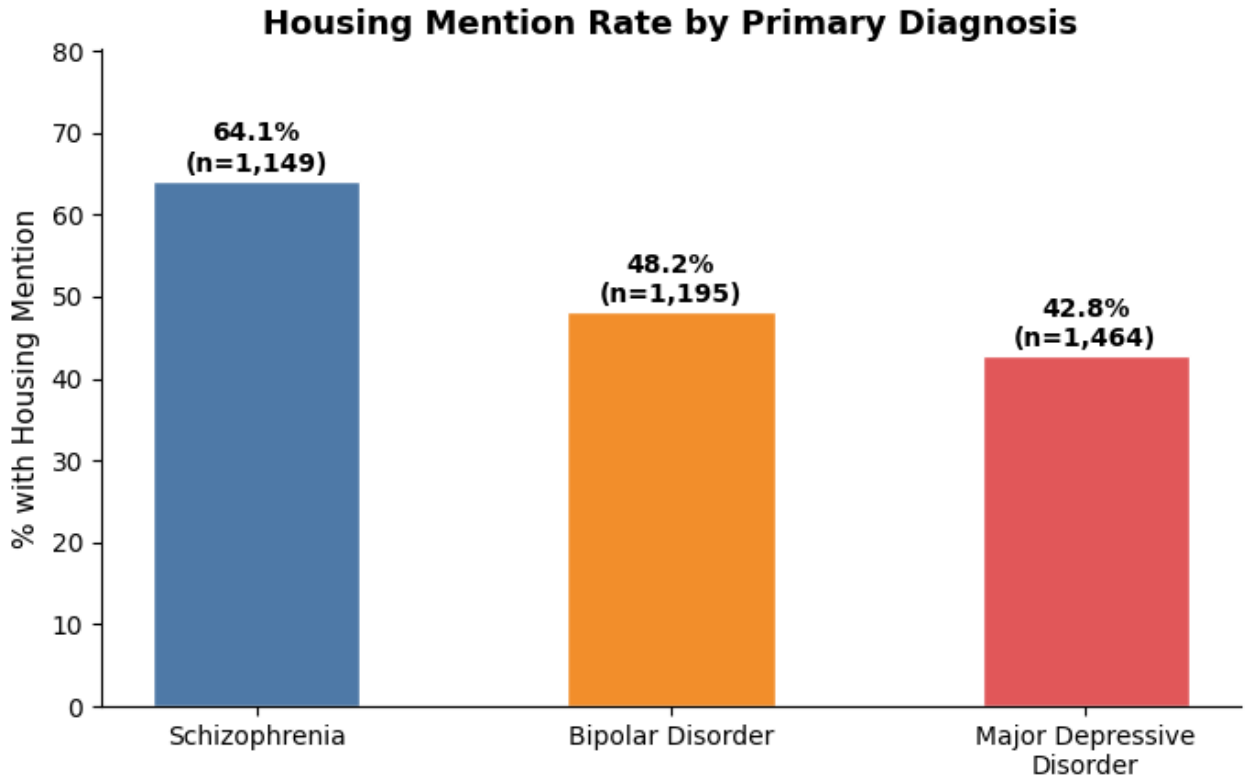


50.9%

of SMI discharge summaries contain a keyword housing mention

- 1,940 of 3,808 admissions flagged
- Likely an undercount given recall of 0.39
- Any gap is clinically concerning here

Schizophrenia leads by 20+ points



64.1%

SCHIZOPHRENIA

48.2%

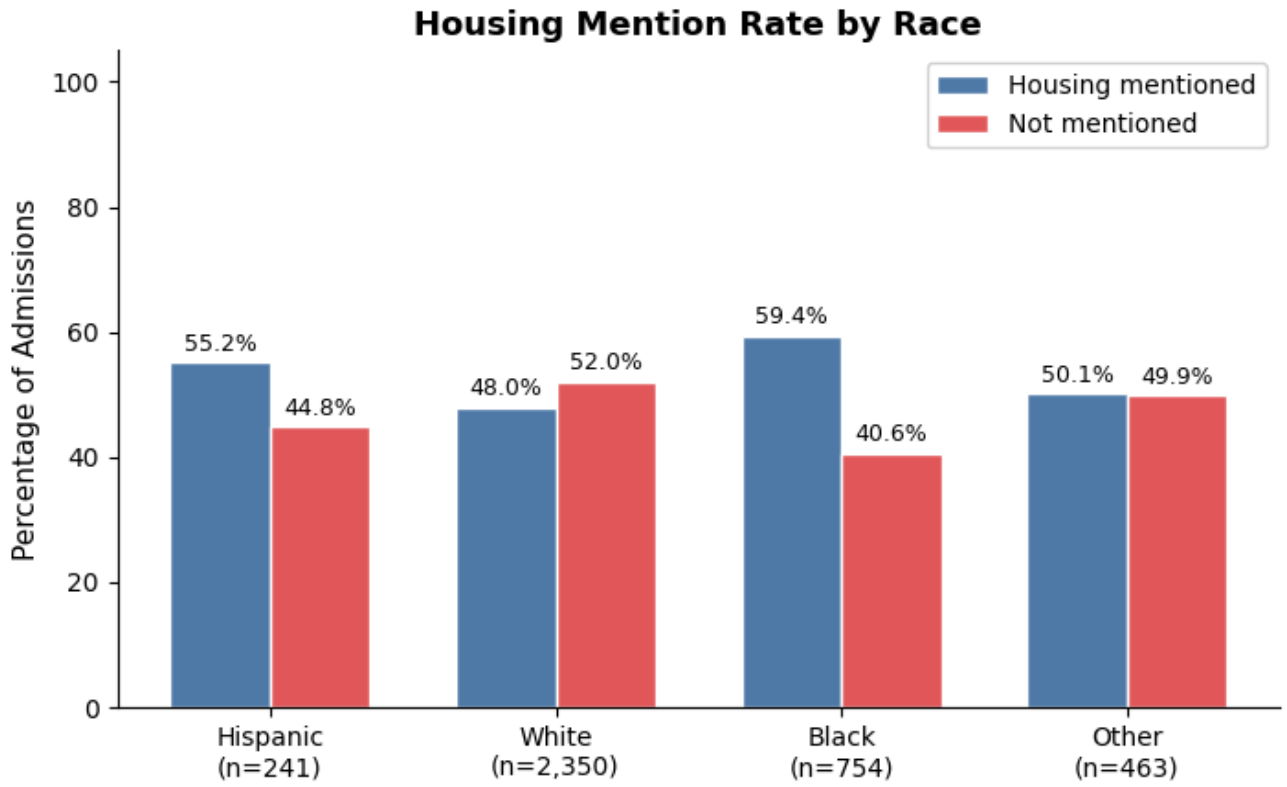
BIPOLAR DISORDER

42.8%

MAJOR DEPRESSIVE DISORDER

Consistent with clinicians asking about living situation more routinely for patients with severe functional impairment.

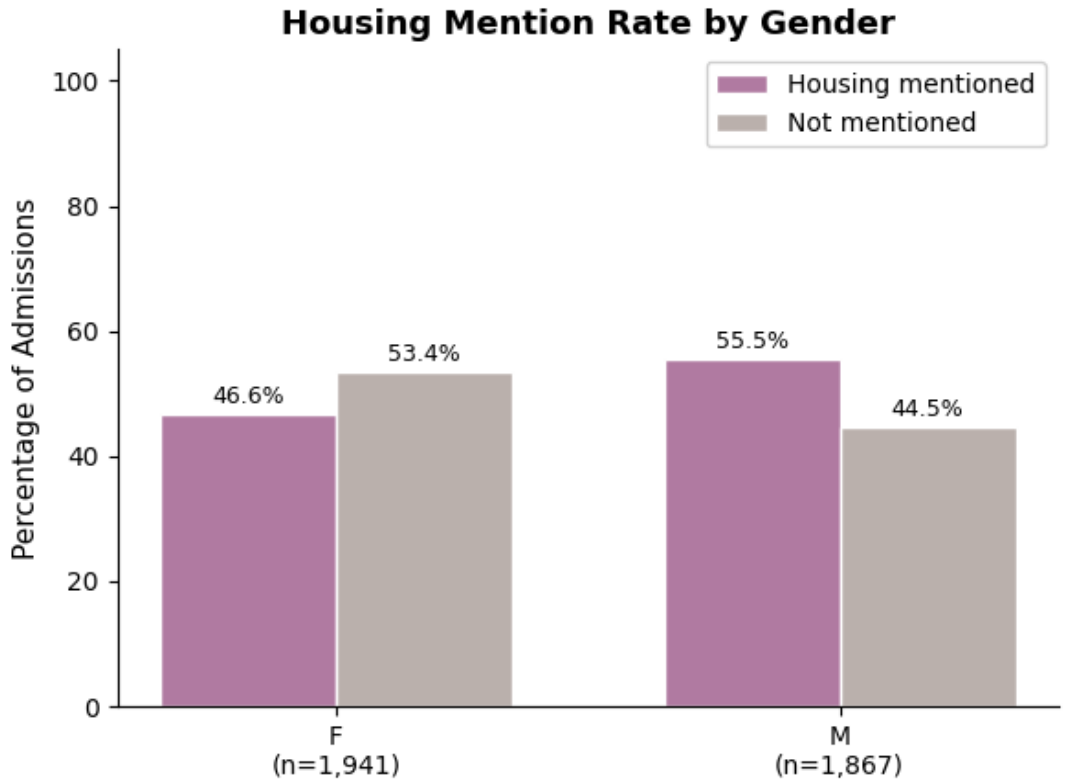
Disparities by race



WHAT THIS COULD MEAN

- Black (59.4%) and Hispanic (55.2%) higher than White (48.0%)
- Could reflect real differences in housing instability
- Could reflect clinician attentiveness to social risk
- White patients may be assumed housing-stable and go unasked

Disparities by gender

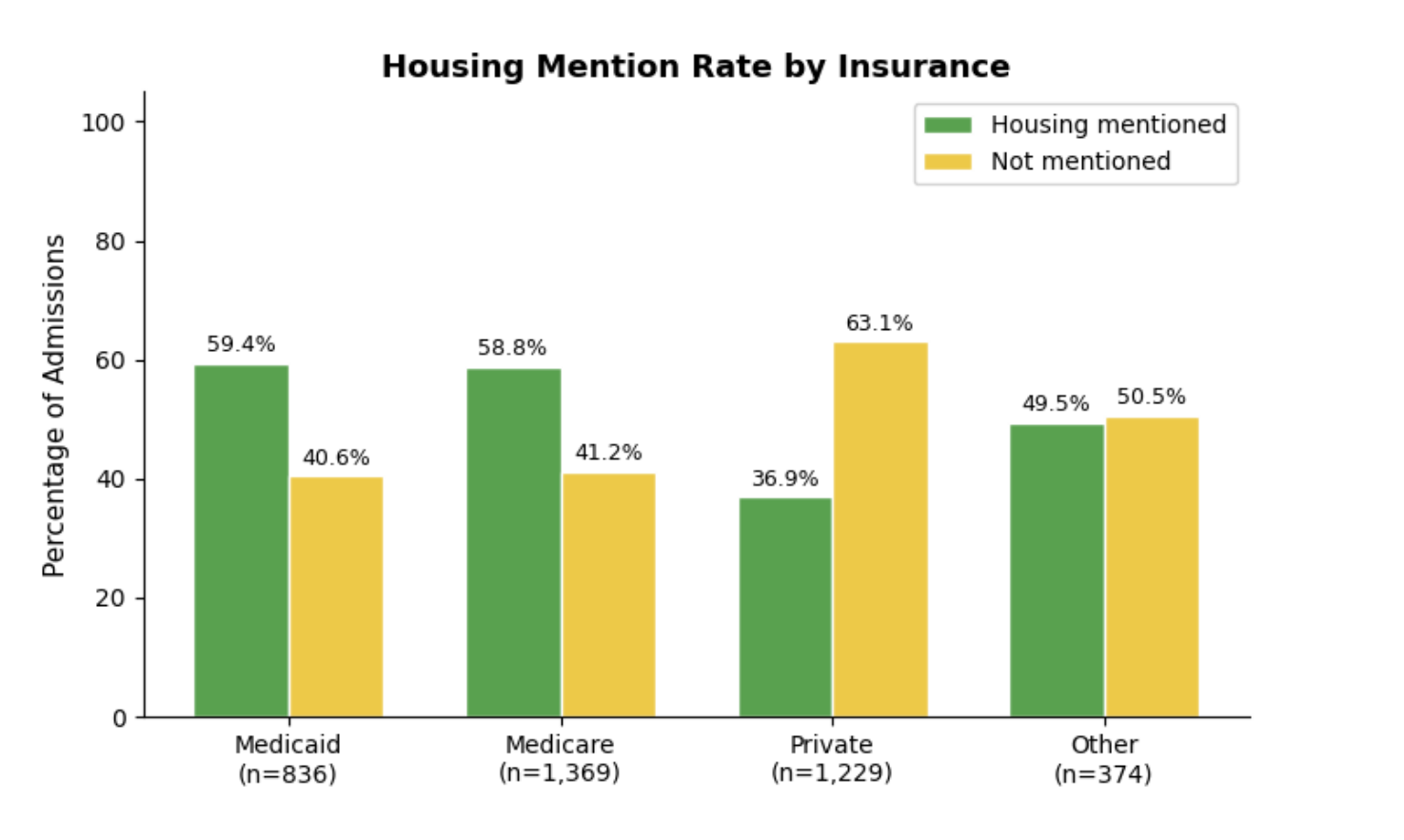


THE 9-POINT GAP

Male 55.5% vs Female 46.6%

- Men are more likely to have housing documented
- But schizophrenia is more common in men — and schizophrenia has the highest mention rate (64%)
- The gender gap may be driven by diagnosis mix, not gendered documentation

The biggest gap is by insurance

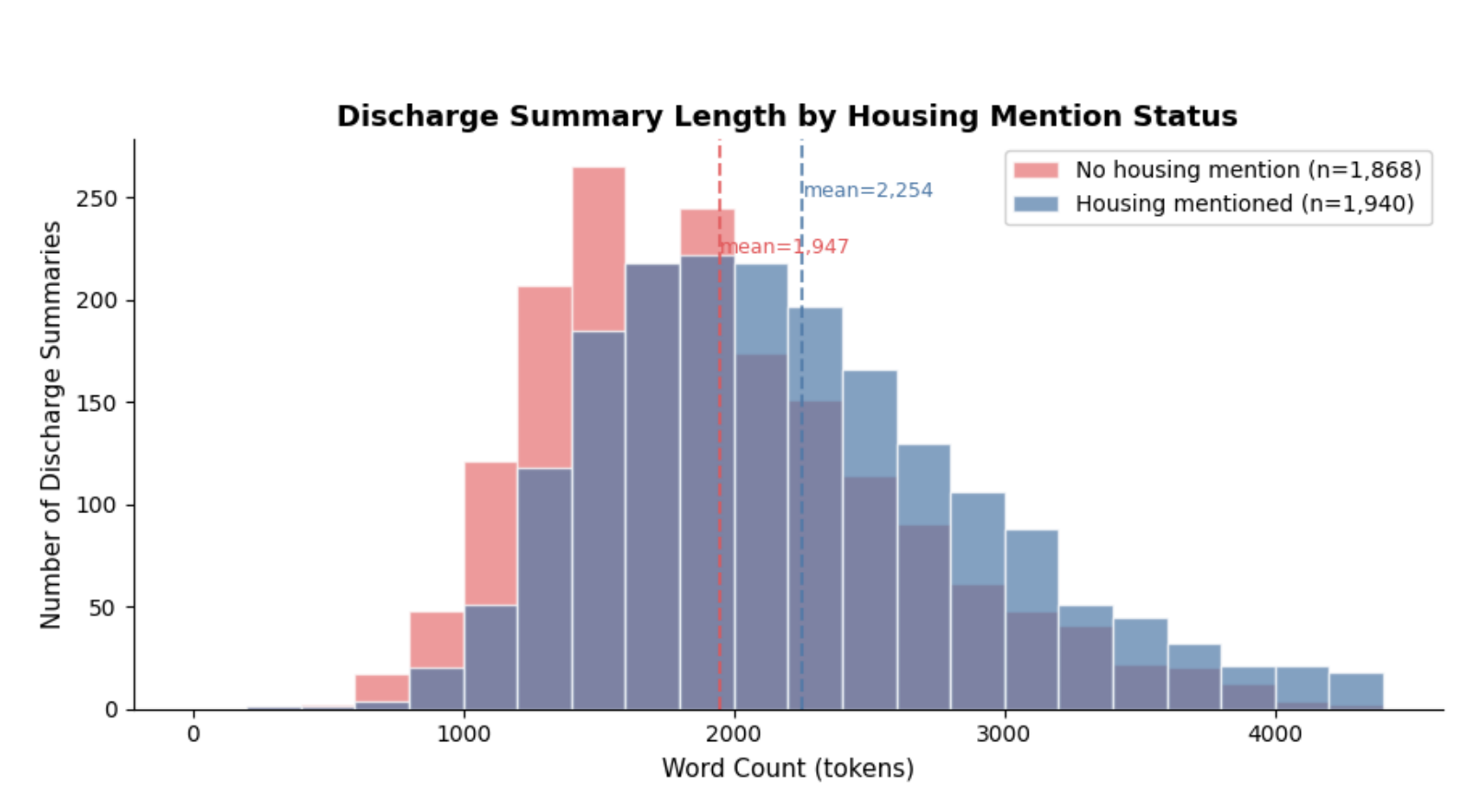


THE 22-POINT GAP

Medicaid 59% vs Private 37%

- Medicaid is income-based — itself a marker of housing vulnerability
- Clinicians may anchor on insurance type when deciding to ask
- Open question: appropriate risk-based attention, or stereotype?

Notes that mention housing run longer



MEAN WORD COUNT

2,254

Housing mentioned

1,947

No mention

A ~300-word gap. Notes that mention housing tend to be more thorough overall — likely capturing more of the patient's social context.

Five things we learned

01 **Half the notes mention housing**

And keyword recall of 0.39 means the true rate is likely higher.

02 **Diagnosis matters more than expected**

Schizophrenia 64% vs MDD 43% — a 21-point gap.

03 **Documentation is unequal across groups**

Black & Hispanic > White; Medicaid & Medicare > Private.

04 **Longer notes mention housing more**

Notes that mention housing tend to be longer — likely because they're more thorough overall

05 **The keyword baseline gives us a floor, not a ceiling**

With recall at 0.39, every rate we report is a lower bound on the true documentation rate

What changed since Milestone 1

M1 — ORIGINAL

How much can a fine-tuned clinical language model improve on keyword rules for detecting social determinants of health?

- Multiple social determinants of health (housing, food, substance use, isolation)
- All discharge summaries in MIMIC-IV-Note
- Hand-label 200–300 notes for evaluation
- Method-focused: keywords vs. ClinicalBERT comparison

M2 — REFINED

Is housing documented in SMI discharge summaries, and do documentation rates differ by race, insurance, or gender?

- Narrowed from all SDOH to housing specifically
- Narrowed from all admissions to the SMI cohort
- Validation set: MIMIC-III-SDOH (1,800 pre-labeled sentences) instead of hand-labeling
- Question-focused: documentation patterns and equity, with the model as the tool to answer it

What happens in Milestone 3

01

Fine-tune

ClinicalBERT on MIMIC-III-SDOH housing labels

02

Apply

Inference on MIMIC-IV-Note SMI discharge summaries

03

Compare

Model vs. keyword baseline; error analysis

04

Re-analyze

Disparity analysis with model labels

Matlin and Brianna will collaborate on every step.

Future considerations

WHAT WE'RE PLANNING

Approach

- ClinicalBERT, fine-tuned on MIMIC-III-SDOH
- Keyword search retained as baseline
- Evaluation against held-out MIMIC-III-SDOH sentences

WHAT'S ON OUR MINDS

Potential Challenges

- Domain shift: training on isolated sentences, applying to full multi-page discharge notes
- How to aggregate sentence-level predictions into a note-level yes/no

WE'D LOVE YOUR INPUT

For our TF

- We're both 209B students, so we need to incorporate an approach or method not covered in class
- Any approaches or methods you'd recommend?

THANK YOU

Questions?

Matlin Gilman · *Brianna Grissom*

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